

Jessica Melani Romero Lora

Complex Systems Engineer & Social Data Scientist

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Interdisciplinary Engineer & Systems Scientist researching Systemic Anomaly Detection and Trustworthy AI. Bridging Complex Systems Theory with advanced Machine Learning (Ensemble methods, Fuzzy Logic) to solve high-stakes problems in Healthcare, InsurTech, and Digital Twins. Ensuring contextual integrity of sensitive user data within fraud detection pipelines. Aiming to contribute to the Trustworthy AI research agenda within the ELLIS network through doctoral research on systemic robustness.

EDUCATION

Masters in AI Engineering (MAEng)

Metropolitan Polytechnic University of Hidalgo
September, 2025 - January, 2027

Masters in Applied Artificial Intelligence (MIA)

Tecnologico de Monterrey, Mexico
Transferred to pursue a specialized Research & Engineering track focused on Mathematical Modeling and Complex Systems
September, 2023 – September, 2024

Social Data Science Essex Summer School Intensive training in Computational Social Science

University of Essex
United Kingdom, April 2022

BSc Sociology and Applied Demography

Specialization in Complexity and Theory Systems and Applied Machine Learning in Social Sciences
Mexico, 2018 - 2022

RESEARCH PROJECTS

Phenotypic Classification of Neurofibromatosis Type 1 (NF1)

Learning Specialization Project | Focus: Machine Learning & Computational Healthcare

- Framework designed to assist clinicians in early phenotype stratification, reducing diagnostic latency using predictive risk-classification models for the early identification of clinical phenotypes in NF1 patients using an Ensemble Learning framework (XGBoost, Random Forest, and Logistic Regression).
- Utilized L1 Regularization (Lasso) and Feature Importance analysis to reduce dimensionality and isolate critical clinical markers.

Robust Fraud Detection in Imbalanced Insurance Ecosystems

Data Engineering Project | Focus: Algorithmic Reliability & Systems Theory

- Architected an XGBoost classifier optimized with SMOTE to effectively manage extreme class imbalance in automated insurance claim systems.
- Engineered dynamic decision thresholds to prioritize Recall, demonstrating a direct impact on preventing systemic financial losses while maintaining model interpretability.

Digital Twin Simulation: EV Battery Survival via Fuzzy Logic

Simulation Lead | Focus: Complex Systems & Decision Theory

- Developed a Fuzzy Inference System to optimize the trade-off between battery longevity and mission completion time in electric vehicles to regulate driving aggression based on non-linear inputs: State of Charge (SoC) and remaining trip distance.
- Implemented rule-based inference (Survival, Eco, and Sport modes) to optimize energy usage, achieving a balance between mission speed and battery longevity.

EXPERIENCE

Data Quality Engineer

March 2024 - Present

Shift Technology (Global AI InsurTech)

Architected privacy-preserving data pipelines compliant with GDPR, directly enhancing model governance for sensitive datasets.

RESEARCH INTERESTS

- AI Safety & Privacy, Contextual Integrity, Socio-technical Systems, Systemic Robustness
- Digital Twins
- Complexity and Systems Theory in Artificial Intelligence

SKILLS

Stack: Python (Scikit-learn, XGBoost, PyTorch/TensorFlow), SQL/NoSQL (JSON, MongoDB), Graph Neural Networks.

Scientific Frameworks: Complex Systems Theory, Applied Demography, Social Data Science (Essex Summer School), and Explainable AI (XAI), Sociological Systems Theory.

Tools & Analytics: IBM SPSS, SAS, Advanced Data Mining, and Scientific Visualization (Matplotlib, Seaborn)